

Armaan Sandhu

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Education

Worcester Polytechnic Institute | Worcester, MA

2025 - Present

B.S. in Computer Science, M.S. in Artificial Intelligence | **GPA: 4.0** | Expected May 2029

Carmel Junior College | Jamshedpur, India | **Valedictorian**

2010 - 2025

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript

Tools: Git, GDB, Linux, Bash, Valgrind, llama.cpp, Redis, PostgreSQL, SQLite

Concepts: OS Kernel Development, Systems & Concurrent Programming, Binary Analysis

AI/LLM: MCP, LLM Routing & Gateways, RAG, Vector Search, Tokenizers, Spec Driven Development

Open Source Contributions

AI Infrastructure (LiteLLM, LanceDB, Lance)

May 2026 - July 2026

- Fixed production data-loss and request-cancellation bugs and added universal gateway support for several AI agent harnesses

Systems & Browser Engines (systemd, CPython, Lightpanda)

May 2026 - Present

- Nearly **20 merged systemd PRs** fixing memory-safety, networking, parsing, and security issues; developed a journalctl optimization delivering **1.7-2.5×** faster reads.

~40 pull requests merged across 6 major open-source repositories, spanning Rust, Python, Zig, and C.

Experience

Founding Engineer & Product Head | MLRouter

June 2026 - July 2026

- Built an OpenAI-compatible gateway for 400+ models with cost/latency routing, provider failover, and enterprise governance; took it from concept to production.

Open Source Author & Maintainer | mcp-persist

May 2026 - Present

- Built and maintain a production-grade MCP persistence layer across Redis, SQLite, and PostgreSQL, reaching **10k PyPI downloads in its first month** and 600+ tests.

CS Peer Learning Assistant, Peer Mentor & Community Advisor | WPI

August 2026 - Present

- Support undergraduate students through foundational computer science coursework, including data structures and systems programming, while providing academic and personal guidance to residential students.

Research Experience

Independent Study, RunixOS | WPI, advised by Prof. Craig Wills

Early 2026 - July 2026

- Built a capability-based OS kernel from scratch, implemented a preemptive scheduler, synchronous / asynchronous IPC, and capability-gated filesystem to study OS design under capability security.

Projects

codebase-rag (Python)

July 2026 - Present

- Built a codebase intelligence tool for AI coding agents that answers cross-file questions in a single query and automatically keeps its index synchronized with code changes.

rag-timetravel (Python)

June 2026 - Present

- Built the core debugging engine behind codebase-rag, letting developers rewind searches to the exact data version used, replay queries, and pinpoint why results changed.

TokenDrift (Python)

June 2026 - Present

- Built a model-upgrade safety check that measures shifts in token counts, costs, and vocabulary before changes reach production, now integrated into MLRouter.

Binary Exploitation & Cache Simulator (C, x86-64 ASM) | WPI, Machine Organization & Assembly

- Reverse-engineered a multi-stage binary and built an N-way set-associative cache simulator with LRU eviction, applying both to validate matrix-transpose optimizations.